1 **(a)** Find a and b such that

$$z^8 - \mathrm{i}z^5 - z^3 + \mathrm{i} = (z^5 - a)(z^3 - b).$$ [1]

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(b) Hence find the roots of

$$z^8 - \mathrm{i}z^5 - z^3 + \mathrm{i} = 0,$$

giving your answers in the form $r\mathrm{e}^{\mathrm{i}\theta}$, where $r > 0$ and $0 \leqslant \theta < 2\pi$. [6]

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